# BOOMIKA S

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## Education

**Bannari Amman Institute of Technology** — **Tamil Nadu, India**
*B.E. Computer Science and Engineering - CGPA: 8.38* — ***2024 – 2028***

**Veveaham Prime Academy** — **Tamil Nadu, India**
*Higher Secondary Education - Score 89.6%* — ***2023 – 2024***

**Greenfield International School** — **Tamil Nadu, India**
*Secondary Education - Score 91%* — ***2021 – 2022***

## Experience

**QiBR Technologies** — **Jun 2026 – Sep 2026**
*DevOps & Infrastructure Intern*

- Engineered containerized microservices deployments using Docker and Docker Compose, integrating Redis, Celery, and Nginx.
- Automated CI/CD workflows using GitHub Actions and AWS Systems Manager (SSM), streamlining application build and deployment processes.
- Established application observability through centralized logging, health monitoring, and deployment runbooks, strengthening operational reliability.

## Projects

**RetailMind AI – Multi-Agent Retail Decision Intelligence Platform** | *React, FastAPI, PostgreSQL (Supabase), LangGraph, Groq LLM, Pyomo, SHAP*

- Built an agentic retail supply-chain platform that forecasts demand quantiles (P10/P50/P90), detects stockout/overstock risk, and recommends interventions via Mixed-Integer Linear Programming (Pyomo + HiGHS).
- Designed a deterministic Python core (forecasting, NetworkX ripple analysis, inventory ledger, dynamic replanning) with two LangGraph/Groq agents for NL analytics, SHAP explanations, and WhatsApp manager approvals.

**AgentQA – Multi-Agent Software Testing & Bug Analysis Platform** | *React, FastAPI, MongoDB, LangGraph, Playwright, Ollama*

- Developed a 9-stage LangGraph pipeline with 8 AI agents that converts SRS documents into test plans and scripts, then runs real Playwright UI tests and live HTTPX API tests with SSRF protection.
- Integrated Bandit/pip-audit security scans, evidence-based Root Cause Analysis (7 failure categories), bounded self-healing retries, and Markdown/JSON/PDF reports; secured with JWT auth.

**SkillQuery – Student Skills Intelligence Assistant** | *FastAPI, PostgreSQL, Groq LLM, WhatsApp Cloud API, JavaScript*

- Built a natural-language assistant that maps skill categories, course aliases, and levels to verified course completions in PostgreSQL, with conversational memory and CSV/Google Sheets export.
- Enforced read-only SQL guardrails (table allowlist, timeouts, row limits) and validated accuracy with a 60-case evaluation suite; served through a web dashboard and WhatsApp.

**InvoiceAI – Handwritten Invoice Processing Platform** | *React, Flask, MongoDB, JWT, Google OAuth, PaddleOCR-VL, LLaMA3*

- Developed a full-stack AI-powered invoice management platform using React, Flask, MongoDB, and REST APIs, implementing PaddleOCR-VL extraction with LLaMA3 post-processing, invoice upload, review, search, filtering, and secure report management.
- Implemented Google OAuth, JWT authentication, role-based access control (RBAC), audit logging, and user/admin authorization to ensure secure multi-user access.
- Integrated the NiBo AI Assistant using Ollama and LLaMA3, and developed analytics dashboards for invoice insights, business analytics, and automated PDF report generation.

## Technical Skills

**Languages**: Java, Python, C, SQL
**Frontend**: React, HTML, CSS, JavaScript
**Backend**: FastAPI, Flask, REST APIs, JWT, Google OAuth, Celery
**Databases**: MongoDB, PostgreSQL, MySQL, Redis
**AI/ML**: Machine Learning, Computer Vision, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangGraph, LangChain, Evals, Ollama, LLaMA3, OCR, SHAP, Optimization, Data Analysis
**DevOps & Tools**: Docker, Docker Compose, Nginx, GitHub Actions, AWS SSM, Kubernetes, CI/CD, Git, GitHub, Linux, Figma
**Core Concepts**: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS)

## Achievements

- **Finalist** – Next SSM Kernel Hackathon 2026, organized by **CDAC**.
- **Finalist** – Snapdragon Multiverse Hackathon 2026, by **Qualcomm**, Bengaluru.
- **2nd Runner-Up** – LudoHack 2025, developed a browser-based game using **p5.js**.
- **Finalist** – NeuroHack, **Shaastra'26, IIT Madras**, built a machine learning solution using the **DREAMER** dataset.
- **Finalist** – Entrepreneurship Hackathon, **STPI × Techniche 2025, IIT Guwahati**.
- **Qualified for the Presentation Round** – Caterpillar Tech Challenge after PPT-based project selection.
- **Solved 370+ problems on LeetCode** with a contest rating of **1555** in Weekly and Biweekly Contests.